Tong-Rui Liu

Computational Mechanics | Scientific AI | Scientific Computing
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Summary

Computational mechanics and scientific AI researcher with a PhD from Imperial College London and expertise in finite element analysis, fracture mechanics, micromechanics, multiscale modeling and GPU-accelerated scientific computing. Research spans physics-based simulation, full-field AI prediction and generative modeling for computational solid mechanics, with experience developing scalable FEM and FFT solvers in JAX, MATLAB and C++.

Core Technical Skills

AI for science: Neural operators, Transformer-based models, Physics-informed machine learning, Surrogate modeling for multiscale mechanics.
Computational mechanics: Nonlinear solid mechanics, Phase-field modeling of fracture, Multiscale & multiphysics modeling, Computational homogenization.
Numerical methods: Finite element method, Virtual element method, Spectral FFT approach, Sparse linear algebra, Self-consistent clustering analysis.
Solver Development: Nonlinear solvers, Bound-constrained optimization frameworks, Iterative linear algebra, Matrix-free methods, FFT-based homogenization.
Programming Language: Python, MATLAB (with MEX), Fortran, C++
Open-source software: JAX, MPI, OpenMP, SuiteSparse, FFTW
CAE and visualization tools: ABAQUS with UEL/UMAT, ANSYS, Gmsh, ParaView.

Professional Experience

•CNRS Laboratory Navier, Institut Polytechnique de Paris *2025 - Now*
Postdoctoral Researcher Paris, France

- Developing nonlinear homogenization and multiscale frameworks for polycrystalline and composite materials, translating heterogeneous microscale behavior into efficient reduced-order constitutive models for large-scale structural simulations.
- Developing a GPU-accelerated JAX-based 3D spectral FFT-based homogenization framework scaling to **200 million** degrees of freedom, enabling full-field strain prediction in heterogeneous microstructures.
- Contributing to an ANR-funded collaborative research programme, coordinating across work packages and working with academic and industrial partners on theoretical developments, computational implementation and project reporting.

Education

•Imperial College London *2020-2024*
PhD in Computational Mechanics London, UK

•Imperial College London, United Kingdom *2018-2019*
MSc in Advanced Computational Methods for Aeronautics, Flow Management and Fluid-Structure Interaction London, UK

•Dalian University of Technology, China *2014-2018*
Bachelor of Engineering in process equipment and control engineering Dalian, China

Research Collaborations and Visiting Appointments

•Research Collaboration with Yale University and University of Pennsylvania *2024 - 2025*
Research Collaborator Remote

- Developed GPU-accelerated JAX-based finite element solvers for dynamic fracture, chemo-mechanical fracture and fatigue, achieving up to 10× speedup over CPU-only implementations.
- Implemented matrix-free Krylov solvers, geometric multigrid (GMG) and bound-constrained nonlinear optimization algorithms to reduce memory requirements and support large-scale fracture simulations.
- Contributed to neural-operator and full-field surrogate modeling for heterogeneous materials, including benchmark development and numerical evaluation in the Micromechanics Transformer (*Micrometer*) framework.
- Contributed to numerical simulation and physical validation for *FunDiff*, a generative-AI framework for physics-governed continuous fields.

•College of Engineering, Peking University *2020 - 2021*
Visiting PhD Student Beijing, China

- Developed reduced-order micromechanical models for 3D woven composites and polycrystalline material using standard and crystal plasticity self-consistent clustering analysis (SCA/CPSCA), achieving up to **two orders-of-magnitude speedup** over traditional numerical methods.
- Built standard and crystal plasticity fast Fourier transformation based homogenization (FFT/CPFFT) for full field mechanical responses prediction using MATLAB-MEX interface with OpenMP.

Selected publications

Journal Articles ([1] for equal contribution, [*] for corresponding author)

- **Tong-Rui Liu[*]**, Jeremy Bleyer, Frederic Legoll, Michel Bornert. *A general data-driven self-consistent clustering analysis for computational homogenization of heterogeneous material with minimal offline stage.* Preprint submitted to Journal of the Mechanics and Physics of Solids. *2026*
- Sifan Wang[1], Zehao Dou[1], Siming Shan, **Tong-Rui Liu**, Lu Lu[*]. *FunDiff: Diffusion Models over Function Spaces for Physics-Informed Generative Modeling.* Nature Communications, vol. 17, 5749. *2026*
- Wuzhe Xu[*], Yulong Lu, Sifan Wang, **Tong-Rui Liu**. *Improving Data Fidelity via Diffusion Model-Based Correction and Super-Resolution.* Journal of Computational Physics, vol. 559, 114883. *2026*
- Sifan Wang[1,*], **Tong-Rui Liu[1]**, Shyam Sankaran, Paris Perdikaris. *Micrometer: Micromechanics transformer for predicting full field mechanical responses of heterogeneous materials.* Computer Methods in Applied Mechanics and Engineering, vol. 448, 118373. *2026*
- **Tong-Rui Liu[*]**, Fadi Aldakheel[*], M.H. Aliabadi. *Hydrogen assisted cracking using an efficient virtual element scheme.* Computer Methods in Applied Mechanics and Engineering, vol. 420, 116708. *2024*
- **Tong-Rui Liu[*]**, Yang Yang, Omar Bacarreza, Shaoqiang Tang, M.H. Aliabadi. *An extended full field self-consistent cluster analysis framework for woven composite.* International Journal of Solids and Structures, vol. 281, 112407. *2023*
- **Tong-Rui Liu[*]**, Fadi Aldakheel[*], M.H. Aliabadi. *Virtual element method for phase field modeling of dynamic fracture.* Computer Methods in Applied Mechanics and Engineering, vol. 411, 116050. *2023*
- Yang Yang, **Tong-Rui Liu**, M.H. Aliabadi, Shaoqiang Tang[*]. *Virtual clustering analysis for long fiber reinforced composites.* Computational Mechanics, vol. 71, page 1139-1159. *2023*

Memberships

- Membership of French Association of Mechanics (AFM) *2026 - Now*
- Membership of European Mechanics society (EuroMech) *2026 - Now*

Talks and International Conferences

- **Conference talk: A general self-consistent clustering analysis for computational homogenization of composite materials using minimal offline stage** *Sep. 2026*
  44th Solid Mechanics Conference: SolMech 2026, Kraków, Poland
- **Invited talk: Virtual element method for computational fracture mechanics** *Jan. 2024*
  School of Mechanics and Aerospace engineering, Dalian University of Technology
- **Conference Talk: Numerical recipes of virtual element method for phase field modeling of brittle fracture** *Sep. 2023*
  21st international conference on fracture and damage mechanics (FDM 2023), London, UK

Awards and Achievements

- **Excellent academic performance scholarship,** Dalian University of Technology *Sep. 2016*
- **National 2nd Prize in Contemporary Undergraduate Mathematical Contest in Modeling,** China Society for Industrial and Applied Mathematics (CSIAM) *Oct. 2015*
- **Outstanding student nomination,** Dalian University of Technology *Sep. 2015*
- **Scientific innovation scholarship,** Dalian University of Technology *Sep. 2015*
- **Excellent academic performance scholarship,** Dalian University of Technology *Sep. 2015*